Machine Learning-Based Drug Repurposing Discovery for Non-Globular MYC and CTNNB1 Oncoproteins

Computer *Engineering Senior Design Project*

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Market & Target Audience

*** Unlocking the 'Undruggable' Oncology Market**

The Problem: Over 70% of cancers involve MYC or CTNNB1, yet they remain "undruggable" due to their disordered structures.
The Solution: An AI-driven pipeline that decodes these targets to identify high-affinity leads from 119 million compounds.

Target Audience:
- **Big Pharma:** Filling oncology pipelines with novel, first-in-class inhibitors.
- **Biotech Hubs:** Accelerating R&D for intrinsically disordered protein (IDP) therapies.

Value Proposition: Reducing initial screening costs by **99%** while providing structurally validated, diverse drug candidates.

Defining the Oncogenic Targets

MYC Proto-Oncogene

Target Name: MYC Proto-Oncogene

PDB ID: 6G6K

Biological Role: Transcription factor; overexpressed in ~70% of cancers

Raw Bioassay Files: ~40 CSV files (PubChem)

Curated Samples: 660 (73% active)

Beta-Catenin (CTNNB1)

Target Name: Beta-Catenin (CTNNB1)

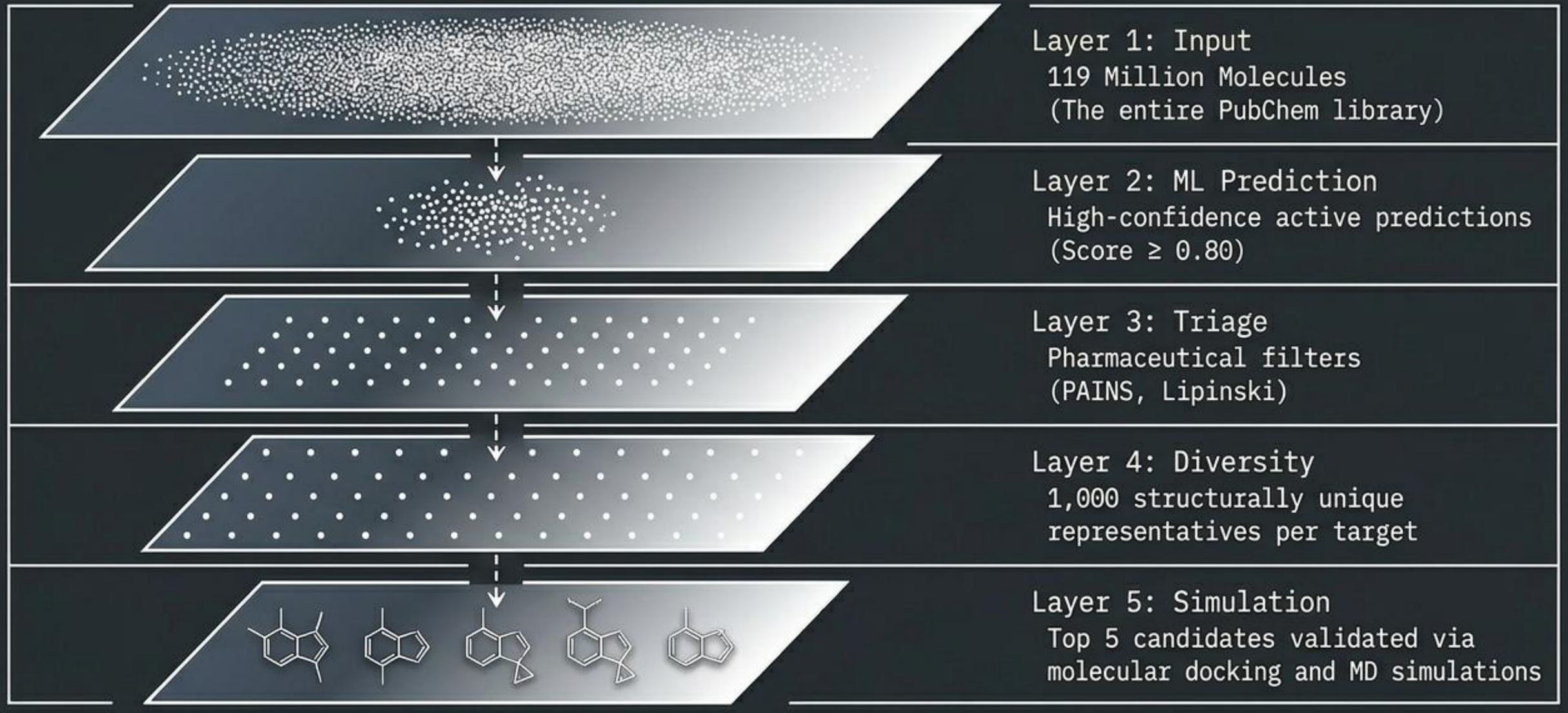
PDB ID: 1JPW

Biological Role: Wnt signalling effector; driver of colorectal & other cancers

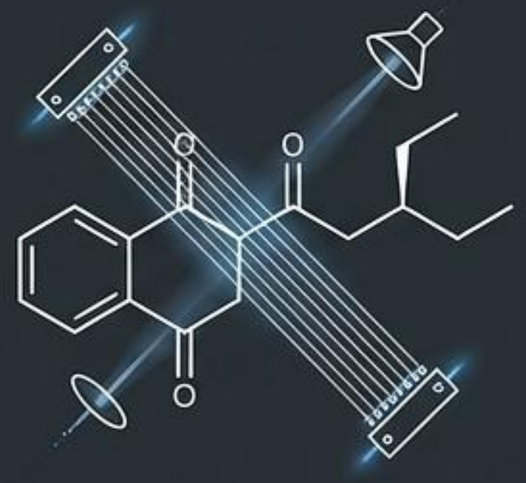
Raw Bioassay Files: ~80 CSV files (PubChem)

Curated Samples: 715 (72% active)

The Scale-Down Architecture



Translating Bioactivity into Binary Vectors



1. Data Curation

Aggregate PubChem BioAssay outcomes into clean datasets.

2. Molecular Fingerprinting

Convert SMILES strings to ECFP4 (Morgan) fingerprints via RDKit (Radius = 2, 2048 bits).

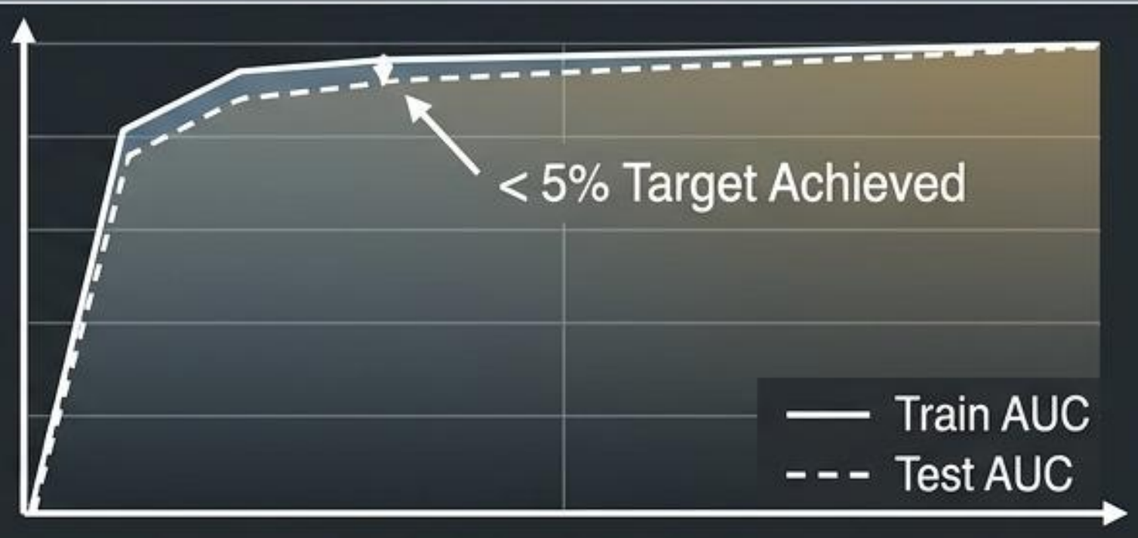
3. Redundancy Elimination

Drop highly correlated features (Pearson r > 0.90) prior to model exposure.

Output Stats: Initial feature space reduced to 1,597 bits for MYC and 1,471 bits for CTNNB1, ready for rigorous selection.

Leakage-Free Feature Selection

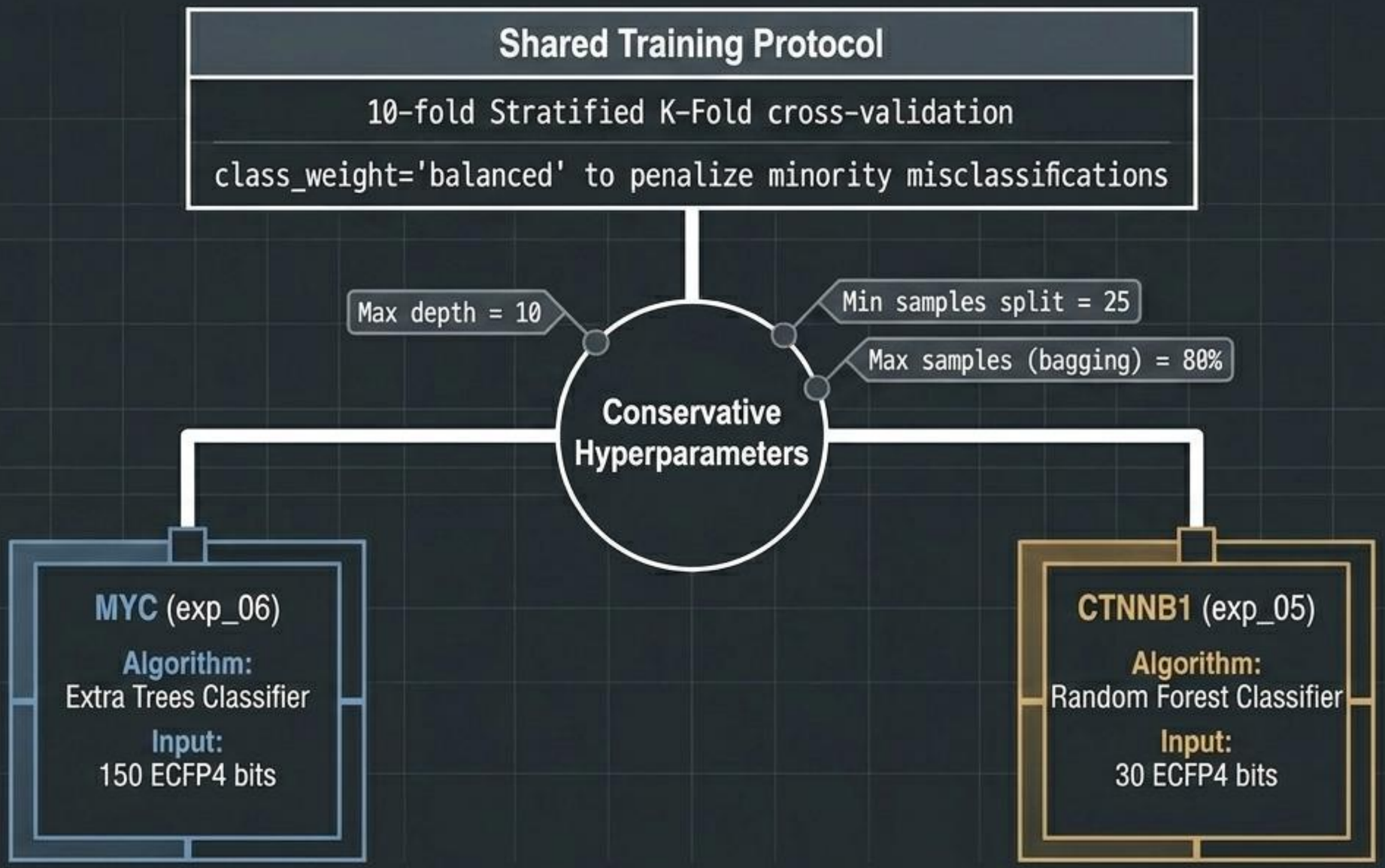
Methodological Rule: Feature importance calculated strictly on the 80% training split. No test data exposure.

Previous Approach (exp_03)	Robust Approach (Optimal K)
	
~315 features	~150 features for MYC / ~30 features for CTNNB1

MYC: 150 features (–52% vs exp_03)
`Train AUC: 1.000 / Test AUC: 0.952`
`Overfitting Gap: 4.75%`
	CTNNB1: 30 features (–90% vs exp_03)
	`Train AUC: 0.987 / Test AUC: 0.939`
	`Overfitting Gap: 4.74%`

Impact: Dramatically reduced model variance while increasing screening velocity by **up to 90%.**

Anti-Overfitting Model Architecture

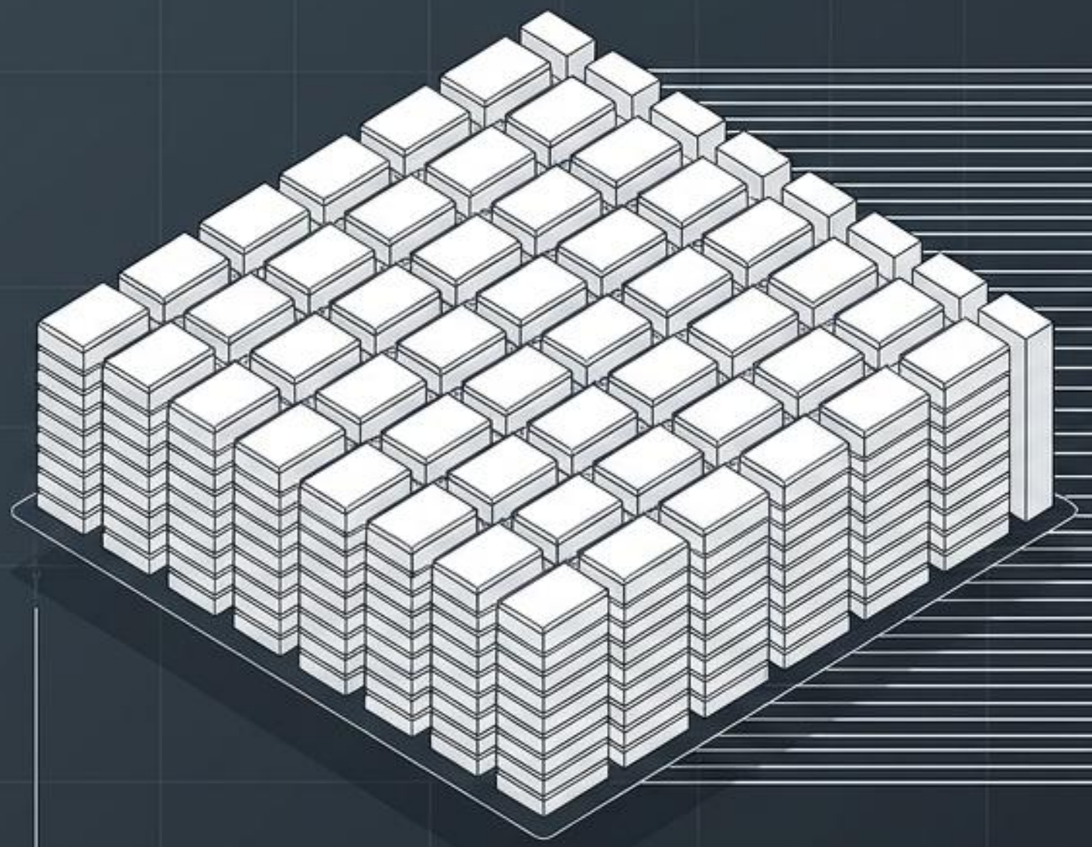


Predictive Performance Validation

Metric	MYC (Extra Trees)	CTNNB1 (Random Forest)
Features	150 ECFP4 bits	30 ECFP4 bits
Test AUC	0.952	0.939
Test F1	0.929	0.909
Test Precision	0.919	0.947
Test Recall	0.938	0.874
Overfitting Gap	4.75%	4.74%

Conclusion: Both .pkl models achieve >0.93 AUC with high precision, ensuring minimal false positives during mass screening.

Massive-Scale Virtual Screening



The Activation Gate (ML Score ≥ 0.80)

Computational Execution: Processed in 60 parallel chunks via joblib with real-time ECFP4-2048 fingerprint generation.

Pharmaceutical Triage and Diversity Clustering

Filter 1: Structural Quality

Applied PAINS and Brenk filters to eliminate pan-assay interference and false positives.

Filter 2: Drug-Likeness

Enforced Lipinski's Rule of Five (≤2 violations) and Veber rules for oral bioavailability.

Filter 3: Chemical Space Coverage

Applied Tanimoto Diversity Clustering (cutoff = 0.60) to avoid selecting minor scaffold variations.

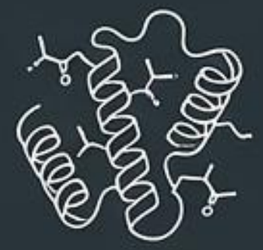
Final Yield: Exactly 1,000 structurally unique, highly viable candidate representatives per target.

Dependency-Free Automated Docking Toolchain

Fully automated pipeline targeting the MYC–MAX protein-protein interaction interface (6G6K).

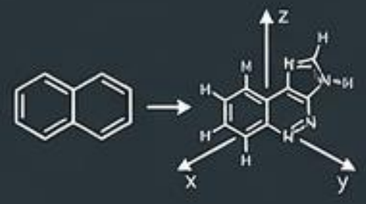
Biopython

Automated protein cleaning and PDBQT receptor isolation.



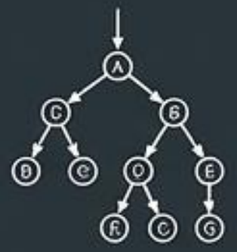
RDKit

Conversion of 2D SMILES into 3D conformers with precise hydrogen addition.



Meeko

Ligand PDBQT preparation (assigning atom types and torsion trees).



AutoDock Vina

Execution via local pre-compiled binaries (no sudo/GUI required).

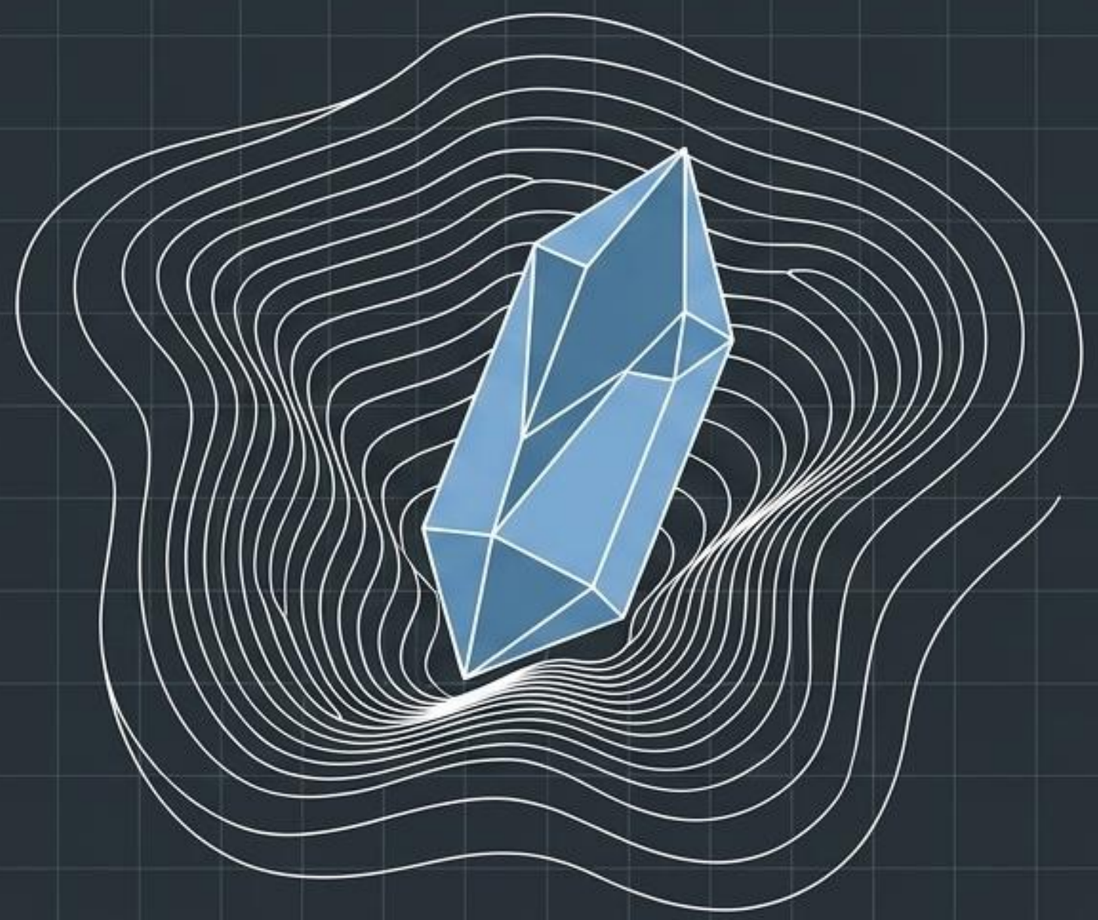
Advantage: Seamless progression from ML candidate directly into 3D binding simulation without manual file manipulation.

Target Binding Results: MYC (PDB: 6G6K)

Top 5 Cluster Representatives (AutoDock Vina)

PubChem CID	Binding Affinity	# Poses
CID 124117970	-7.56 kcal/mol	9 Poses
CID 137355896	-7.42 kcal/mol	9 Poses
CID 155548927	-6.98 kcal/mol	9 Poses
CID 46206285	-6.76 kcal/mol	9 Poses
CID 91515241	-6.42 kcal/mol	9 Poses

Insight: All final simulated candidates successfully demonstrate strong binding affinities within the −6.4 to −7.6 kcal/mol range at the critical MYC–MAX interface.



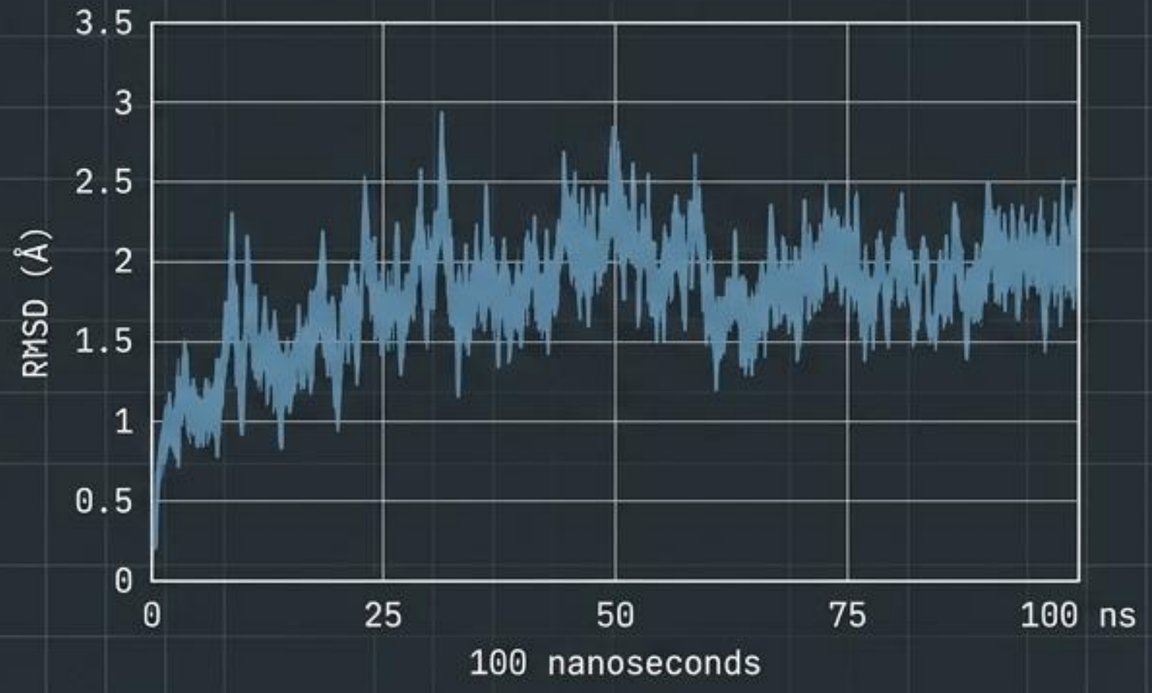
Dynamic Validation via MD Simulations

Outcome: MD trajectories will dictate the final, definitive ranking of the candidates.

Limitation: Static Binding (AutoDock Vina)

AutoDock Vina provides a static snapshot of binding affinity.

Current Execution: Dynamic Stability (MD Simulations)



Objective: To measure dynamic binding stability, track RMSD over time, and confirm the ligand remains securely locked in the MYC–MAX interface under physiological conditions.

Interactive Deployment: The Web Portal

Protein Selector

Tab-based navigation between MYC and CTNNB1 datasets.

Export Ready

Direct CSV export functionality to enable immediate lab testing.

Interactive Analytics

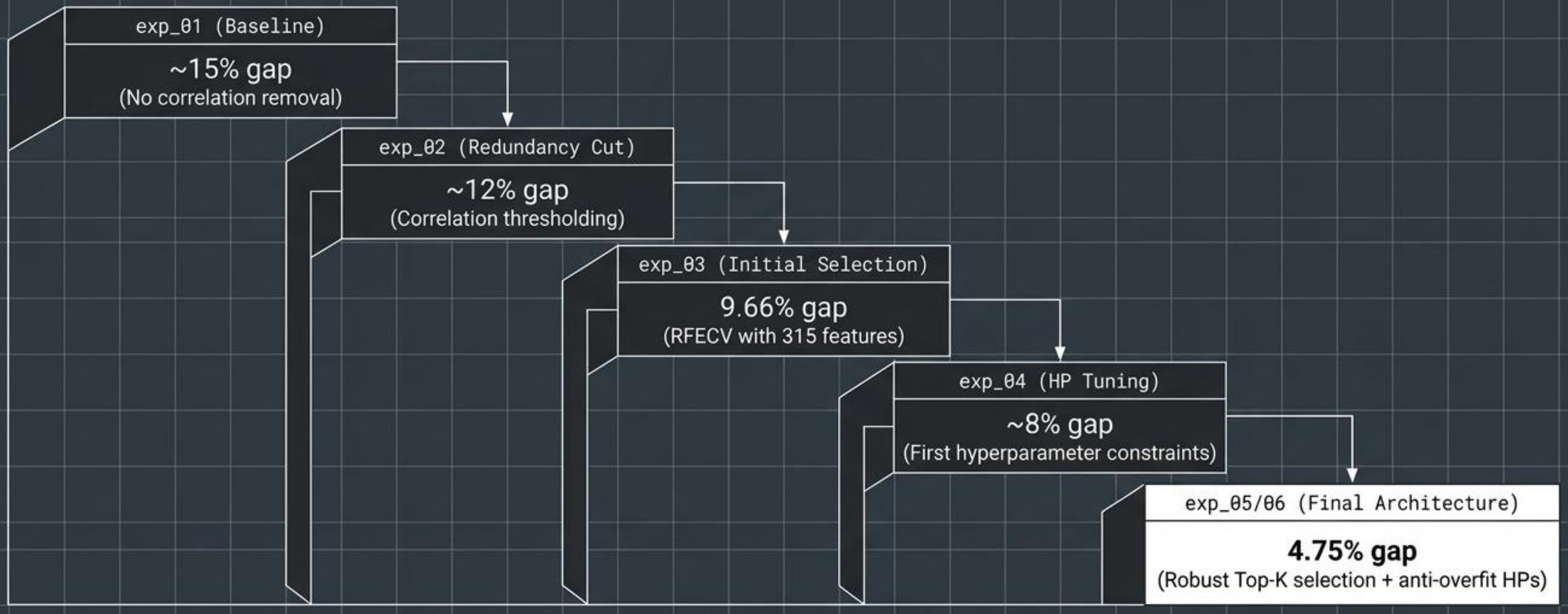
Plotly scatter plots comparing ML scores against molecular weight, logP, and TPSA.

Candidate Triage

Sortable tables featuring CIDs, SMILES, and Lipinski flags.

Platform: Streamlit-based visualization dashboard built with an Inter-inspired academic dark-mode UI.

Methodological Evolution & Impact



Synthesis: By treating feature selection and generalization as the primary engineering constraints, the pipeline evolved from a standard ML script into a rigorous, leakage-free discovery engine capable of scanning 119M compounds safely.

NotebookLM

Strategic Trajectory and Next Steps

Project Objective Maintained:
Delivering open, reproducible, high-precision hit generation for hard-to-drug oncogenic targets.

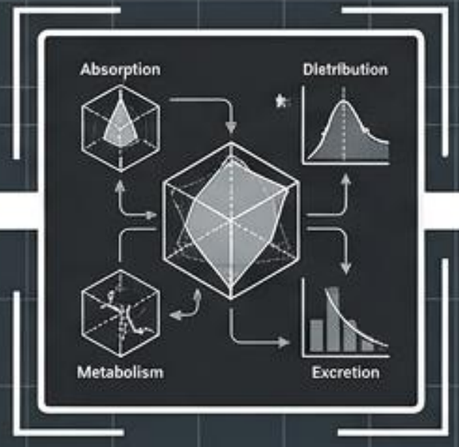
1. Trajectory Integration

Complete 100 ns MD simulations and merge dynamic stability metrics into the final MYC candidate rankings.

2. Target Expansion

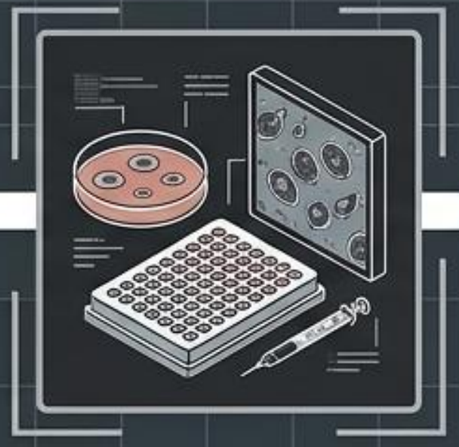
Execute the automated AutoDock Vina pipeline for the top 5 CTNNB1 cluster representatives.

3. Pharmacokinetic Profiling



Conduct advanced ADMET computational profiling for the finalized candidate lists.

4. Experimental Validation



Transition the top 10 computationally validated compounds toward in-vitro biological testing.

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THANKS ..

Explore the Results:
Portal: http://172.16.10.190:8502
Access: The portal is currently hosted on the university's research server for data security. (Accessible via MSKU Campus Network.)